



THE UNITED REPUBLIC OF TANZANIA
MINISTRY OF DEFENCE AND NATIONAL SERVICE
MILITARY SCHOOLS ASSOCIATION
FORM TWO PRE NATIONAL ASSESSMENT



031

PHYSICS

TIME: 2:30 Hours

GEPAM science hub

WhatsApp 0748978398

07th AUGUST 2026

INSTRUCTIONS

- 1 . This paper consists of section A , B and C with total of ten (10) questions .
- 2 . Answer ALL the questions .
- 3 . All writing must be in blue or black ink , except drawing which must be in pencil .
- 4 . Communication devices and any unauthorized materials are not allowed in the assessment room .
- 5 . Write your assessment number at the top right corner of every page .
- 6 . Where necessary the following constants may be used .
 - (i) Acceleration due to gravity , $g = 10 \text{ m/s}^2$
 - (ii) Density of water $\rho_w = 1 \text{ g/cm}^3$ or 1000 Kg/m^3

Question number	For Assessor's use only	
	Score	Assessor's initial
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
Total		

SECTION A (15 Marks)

Answer all questions in this section .

1 . For each of the items (i) - (x) , choose the correct answer among the given alternatives and write it's letter in the box provided .

(i) Why does a hot air balloon rises into the air ?

- A . The balloon becomes heavier
- B . The air inside the balloon becomes less dense than the surrounding air
- C . The balloon is denser than the surrounding air
- D . The balloon is pulled upward by the wind .

(ii) A body moving with constant velocity has :-

- A . Zero acceleration
- B . Increasing acceleration
- C . Negative acceleration
- D . Variable speed .

(iii) Which of the following is an advantage of secondary cells ?

- A . They cannot be recharged
- B . They always produce a higher voltage than primary cell
- C . They are used only once
- D . They can be used repeatedly after recharging .

(iv) Light rays refracted from the convex mirror appear to :-

- A . Coverage at the focus
- B . Diverge from the focus
- C . Stop at the pole
- D . Travel parallel to principal axis .

(v) A spring extends by 0.20 m when a force of 10 N is applied . What is it's spring constants :-

- A . 20 N/m
- B . 50 N/m
- C . 100 N/m
- D . 200 N/m

(vi) Which of the following instruments is most suitable for measuring internal diameter of 100 ml beaker ?

- A . Metre rule
- B . Vernier caliper
- C . Measuring tape
- D . External caliper

(vii) A falling object is pulled down by the earth .The earth is pulled towards the objects

why doesn't the earth move ?

- A . Because the earth has only gravity
- B . Because the earth has a very large mass and small acceleration
- C . Because the earth has a very small and large acceleration
- D . Because air resistance gets in the way .

(viii) A student lowers an object into a liquid filled in a container . He finds that there is a maximum apparent loss in weight of the object when :-

- A . It just touches the surface of liquid
- B . It is just partially immersed and also touches the sides of the container
- C . It is completely immersed in the liquid
- D . It is partially immersed in the liquid

(ix) The sea water is denser than fresh water due to :-

- A . Mixing of sand
- B . Stagnation
- C . Mixing salts
- D . Evaporation .

(x) A student want to compare the densities of two different liquids . Which instruments should the students use for a direct comparison ?

- A . Barometer
- B . Thermometer
- C . Hygrometer
- D . Hydrometer .

Answers

i	ii	iii	iv	v	vi	vii	viii	ix	x

- 2 . Match the descriptions of the terms used in magnetism in list A with its corresponding concept used in magnetism in list B by writing the letter of the correct response beside the corresponding item number in the box provided .

List A	List B
(i) A location where the total magnetic field equal to zero	A . Magnetisation
(ii) The process of aligning the atomic dipoles in one direction	B . Demagnetisation
(iii) The region around the magnet in which magnetic force can be experienced by magnetic materials	C . A neutral point
(iv) The direct current supplied in the wire to produce magnet	D . Earth magnetic
(v) The process of limiting the flow of magnetic field across an object	E . Magnetic shield
	F . Magnetic field
	G . Stroking
	H . Electromagnetic

Answers

List A	i	ii	iii	iv	v
List B					

SECTION B (70 Marks)

Answer all questions from this section .

- 3 . (a) From the concept of Physics , we see that weight of an object is equal to the gravitational force on it . List two properties of gravitation force .
- (i)
- (ii)
- (b) Assume a fixed mass rocket that moves from the earth to planet X . If it weighs 10,000 N on the earth and 300 N on planet Y . Determine the acceleration due to gravity on the planet .
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Candidate's Examination Number

4 . (a) A car starts from rest and is accelerated uniformly at rate of 4 m/s^2 for 5 sec . It maintains a constant speed for 20 seconds , brakes applied and then stops in the next 3 seconds . Find :-

(i) Draw a velocity – time graph of the motion .

(ii) Find the total distance travelled .

(b) A box weight 100 N and has a volume of $2 \text{ m} \times 4 \text{ m} \times 6 \text{ m}$. What greatest pressure can be exerted by a box ?

5 . (a) Explain the meaning of the following concepts .

(i) Opaque objects

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(ii) Translucent object

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(iii) Transparent objects

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Candidate's Examination Number

(iv) Luminous objects

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(b) (i) State laws of reflection of light .

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(ii) Calculate the number of images when plane mirrors are inclined at 45° .

6 . (a) Explain the following observation as applied in Physics .

(i) Hot soap is tastier than cold soap

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(ii) Ink sticks on paper

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(iii) Kerosene rise through the wick of lamp

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(iv) Pond skater man to walk on the surface of water .

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(b) A girl of mass 50 Kg walks up a flight of 12 steps . If each step is 30 cm high , calculate the work done by the girl climbing the stairs .

7 . (a) During a Physics lesson at Ruhuwiko Secondary School , Form Two students were discussing the topic of Density . Some students argued that leaning density has no importance in real life and that it is only a theory in the classroom . As the physicist explain to the students the importance of studying density by giving three examples of its applications in daily life , science and technology .

(i)

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(ii)

(iii)

(b) A physics student wants to determine the relative density of kerosene . The object has a weight of 8.0 N in air , 6.0 N when fully immersed in water , and 6.4 N when fully immersed in kerosene . Calculate the relative density of kerosene .

8 . (a) Mention three factors on which buoyant force / upthrust depends .

(i)

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(ii)

(iii)

(b) A piece of cork of volume 100 cm^3 and density of 0.25 g/cm^3 is floating on water .
Calculate the :-

(i) Mass of the cork

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Candidate's Examination Number

(ii) Volume of the cork immersed in water

9 . (a) Briefly explain how the capacitor becomes charged .

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(b) Briefly explain how the steal bar can be magnetised by electricity .

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SECTION C (15 Marks)

10 . (a) Two resistors are connected in series and parallel separately . Which connection is commonly used in household wiring and why ?

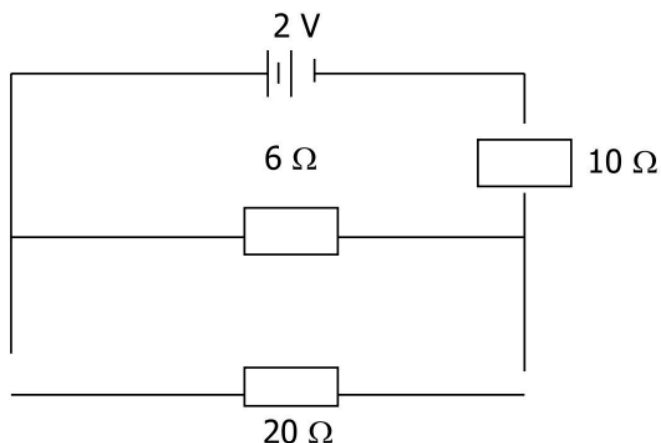
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(b) The battery in the circuit shown in the following diagram has an e.m.f of 2 V ad negligible internal resistance .



What will be the current flowing in $6\ \Omega$?

(c) A circuit in a house is protected by 10 A fuse . The circuit is connected to the 240 V mains . The following appliances are connected to the circuit .

Appliance	Power rating
Bulb 1	100 W
Bulb 2	75 W
TV	300 W
Heater	1500 W

Determine whether the fuse will blow Off or ON . If all appliances are turned on .